MASA CHEUNG

(646) 289-2641 cheung.masa@gmail.com **PORTFOLIO** **GITHUB** **LINKEDIN**

SKILLS

JavaScript, React, Redux, PostgreSQL, Docker, Git, HTML5, jQuery, Test Driven Development (TDD), Prometheus, Grafana, WebSockets

EDUCATION

BA Computer Science - *Rutgers University - New Brunswick*

EXPERIENCE

BGC Partners, **Software Engineer** Feb 2022 - Present

Credit Match & Quick History

- Owned Quick History's rewrite from Java to React and Redux, implementing APIs, SQL queries, and Credit Match schema mappings for trade and order history; eliminated the Market Data dependency and delivered to production within four months.
- Delivered SEF/DTO access overrides in Java for a major trading desk within three days, then introduced an Oracle configuration table in the DataDepot schema and cross-schema Credit Match entitlement checks, enabling Product Support to manage buyer and seller permissions by business, desk, and product.

Product Builder & Volume Match & Event Service

- Partnered with trading desks and project managers to translate requirements and feedback into reusable Vue.js and Backbone.js components and JSON-driven matrix, triangular, and list layout builders, enabling Volume Match to expand from 2 to 7 desks across the UK and US and from 6 to 22 layouts over three years.
- Enabled spread and butterfly creation during live auctions through Vue.js drag-and-drop and search workflows, integrating WebSocket updates, instrument naming, price calculation, and buy/sell order submission to replace advance setup by desk or static-data teams.
- Owned Product Builder's React development and delivery, coordinating sprint priorities across distributed teams while implementing dynamic forms with reusable components and integrating instrument-enrichment services for storage in Credit Match's Oracle database.
- Implemented a reusable API for weighted and unweighted spreads and butterflies, defining WebSocket message contracts with backend engineers and linking each leg to its underlying outright instrument for downstream pricing.
- Automated testing, Docker image publishing to Nexus, and Docker Compose deployments through GitLab CI/CD across eight development/QA environments, beta, and production, enabling single-click releases and rollback to prior versions while restricting production deployments to Git tags.
- Engineered a Dockerized Node.js and Express monitoring service to capture user events and trade round-trip latency, integrating Prometheus metrics, PostgreSQL event logs, and Grafana dashboards and failure alerts to reduce manual monitoring by one hour per day.
- Diagnosed missing instruments in Volume Match trading grids, collaborating with trading desks and tracing WebSocket auction messages, static data, pricing inputs, and UI rendering through Grafana and cross-application logs; documented reproduction steps in Jira and created wiki runbooks for recurring-issue troubleshooting.

PROJECTS

MasaNote (Rails, React.js, Redux.js, ReactQuill, JavaScript, PostgreSQL, SCSS) github

MasaNote is a single-page note-taking app that features a rich-text-editor, image uploading and autosaving

- Engineered secure authentication and session management using BCrypt password encryption, RESTful APIs, and cryptographically secure token generation with caching, strengthening application security and user access controls.
- Built key user-facing features using React, Redux, HTML, and SCSS, including a customized React-Quill editor with auto-save, Notebook management, and scalable Redux state management.